Reaction Time Experiment — Feedback and Worked Example

A Course on Data Visualization · Commentary on the Group Activity

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How to use this feedback

Compare this commentary with your group's report and the decisions recorded in your analysis plan. It describes features of a convincing analysis and provides one worked approach. Other approaches can be equally appropriate when their questions, summaries, and interpretations are consistent.

This is general feedback on the activity, not an assessment of particular submissions. No student reports or observations from `RT_data.csv` were available when this document was prepared. The example therefore generates its numerical results from the CSV when it is run; no findings are supplied in advance.

The planned design has 24 participants, two conditions (A = easy, B = hard), and 40 trials per participant per condition. These are design expectations, not verified counts. The visual-task scenario accompanying the activity is fictional and must be identified as such in a report.

> **The key question for reviewing your work:** do the data selected, the unit of summary, the graph, and the written conclusion all refer to the same question?

Critical issues to discuss

Repeated measurements and the unit of analysis

A plot of all trials describes trial-level variability. A plot of participant summaries describes variation between participants. Specify which level each output represents. If you calculate a within-participant difference, define its direction explicitly, for example B minus A, and keep each participant's values matched across conditions.

The planned 1,920 rows are not 1,920 independent participants. Pooling trials can be useful for describing their distribution, but it conceals pairing and can give participants with more eligible trials greater influence. A graph connecting each participant's A and B summaries makes that pairing visible. A separate graph or table of differences addresses how consistently the within-participant change occurs.

For example, let m_{ic} be participant i's median reaction time on eligible trials in condition c. Then

$$\Delta_i^{RT} = m_{iB} - m_{iA}.$$

A positive difference means a longer median reaction time in B. The mean of these participant differences is a different summary from the difference between the two pooled trial medians. Label the quantity you actually calculate. Similarly, the median of paired differences need not equal the difference between the two medians of participant summaries.

Reaction time and correctness

Decide whether the main reaction-time analysis uses all responses, correct responses only, or separate summaries for correct and incorrect responses. Explain how that choice changes the question. If you analyse reaction time on correct trials only, calculate accuracy from the relevant set of all trials, not from the correct-only subset. Identify participants with no eligible trials rather than silently dropping them.

For participant i in condition c, the observed-response accuracy is

$$\widehat{p}_{ic} = \frac{\text{number of correct responses}_{ic}}{\text{number of responses with observed correctness}_{ic}}.$$

Using only correct trials in both numerator and denominator produces 100% accuracy by construction. Excluding a reaction time from the RT analysis does not automatically justify excluding its observed correctness from the accuracy analysis.

A correct-only RT comparison is conditional on success. If the conditions have different error rates, the retained trials may represent different subsets of performance. Report accuracy alongside reaction time. Slower responses accompanied by greater accuracy may be consistent with a speed–accuracy trade-off, but those descriptive patterns alone do not establish its mechanism.

Distribution shape and unusual values

Inspect the distribution before choosing measures of centre and spread. Consider what the mean and standard deviation reveal compared with the median and interquartile range. A boxplot outlier flag alone is not a reason to remove an observation. Any exclusion rule needs a justification and the number of affected trials and participants. For discretionary exclusions, check whether the main description changes when the rule is omitted.

Distinguish impossible or unusable recorded values from unusual but potentially valid observations. The worked example excludes missing, non-finite, and non-positive RT values from RT summaries. It retains all other positive finite RTs, including extreme values, because no verified timing protocol is available to justify an additional cutoff. This choice is transparent, not a claim that every retained value is plausible.

Missingness, denominators, and weighting

Report how many observations contribute to each result. Distinguish a pooled trial proportion from the mean of participant-specific proportions: the pooled calculation weights participants equally only when their denominators are equal. The two estimates can also happen to coincide with unequal denominators, but their weighting is still different.

Exclusions can make an initially balanced design unbalanced. If correctness is missing, state whether the denominator includes only trials with observed correctness and report how many values are missing. This denominator rule does not remove possible bias from missingness.

Report the number of complete participant pairs separately for each outcome. A participant may contribute to the paired accuracy comparison but lack eligible RTs in one condition. Also distinguish an absent participant–condition combination from a combination containing trials but no usable outcome.

Visual choices

Use common axes when comparing groups, show units and sample sizes where useful, and distinguish trial-level from participant-level variability. Explain choices such as histogram bin width, density bandwidth, or

a transformed axis when they affect interpretation. Make condition labels readable and avoid relying on colour alone.

A bar chart of mean RTs hides the shape of the distribution and the within-participant pairing. Histograms, empirical distribution functions, or boxplots with observations address distributional questions; connected participant summaries address paired comparisons. Neither graph replaces the other.

If error bars are included, define whether they show a standard deviation, standard error, or interval estimate, and explain the unit on which they are calculated. Trial-level variability and uncertainty about a participant-level average are different quantities. The worked example uses descriptive plots without inferential error bars.

Scope of the conclusions

Separate an observed difference from a claim about why it occurs. Do not infer causation or actual randomization from the CSV or from the fictional scenario. This activity focuses on exploration and description; formal hypothesis tests and model fitting are not required.

A smaller descriptive mean is not automatically a statistically significant difference. Conversely, avoiding a significance test does not prevent you from describing the magnitude, direction, and variability of an observed paired difference. Do not interpret failure to see a clear pattern as proof that the conditions are equivalent.

Further advice for a stronger report

Make the chain from question to conclusion visible

A useful sequence is **question → observations included → summary → graph or table → interpretation → limitation**. A reader should be able to follow that sequence without inspecting every line of code.

Issue to look for in your report	A useful revision
"B is slower" with no definition	State which RT summary is compared, on which trials, and whether the comparison is paired.
Only a group average is reported	Add a distributional view and participant-level variation.
Accuracy changes after filtering out slow RTs	Explain the accuracy population and justify any RT-based restriction; do not inherit it accidentally.
Every unusual value is removed	Separate invalid measurements from unusual observations and justify any discretionary rule.
Counts differ between figures	Identify the filters, missingness, and contributing participants for each figure.
An AI explanation mentions results absent from the output	Check each claim against a computed value or visible feature and remove unsupported claims.
Two groups disagree	First align their units, trial subsets, denominators, and summaries; then recompute the disputed quantity.

Keep numerical precision sensible and round only for display. An accuracy difference of 0.05 is **5 percentage points**; it is not necessarily a 5% relative increase. If you use a logarithmic axis, distinguish changing the display scale from analysing log-transformed times.

Verify the AI-assisted work

A strong prompt helps, but it does not validate its response. Recalculate at least one participant's RT summary and accuracy directly, inspect the participants lost from paired comparisons, and check warnings and missing outputs. Inspect the CSV's coding rather than assuming that a plausible variable name establishes its meaning.

Keep the original prompt and relevant revisions. Record what was accepted, changed, or rejected and why. A reproducible report should run in a fresh R session and explain the origin of the data and the role of AI assistance.

A worked example of prompting

The prompt below makes concrete analytical choices rather than asking generically to "analyse the data". It specifies the question, repeated measurements, outcome definitions, validation, plots, and desired output. Its choices form one reasonable plan, not a compulsory solution.

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Create one complete R Markdown report, in English, for a bachelor-level
Data Visualization activity. Return its full .Rmd source, including YAML,
explanations, and executable R chunks. Use tidyverse and knitr.

The input is data/RT_data.csv. I am not supplying its observations in this
prompt, so do not invent findings or silently simulate replacement data.
Generate numerical results from the file when the report is knitted.

Planned design: 24 participants, each completing 40 trials in A (easy)
and 40 in B (hard). Every row is a trial; measurements repeat within
Subject and the conditions are paired within participants. The associated
visual decision task is a fictional teaching scenario, not verified
documentation of the CSV's provenance. Verify actual counts in the file.

Columns: Subject = participant ID; Condition = A/B; RT = milliseconds;
Correct = numeric 0/1 (incorrect/correct); Correct_f = text labels.
Preserve the imported data. Check required columns, parsing problems,
identifiers, condition labels, and the 0/1 coding. Show a cross-tabulation
of Correct and Correct_f for manual checking. Stop on invalid identifiers
or coding rather than guessing repairs. Do not remove repeated rows merely
because their recorded values are identical.

Question: how do reaction time and accuracy differ between conditions,
and how consistent are the differences across participants?

Use the following analysis plan:
1. Audit the observed design and missingness. Display a participant-by-
   condition table with trial counts and outcome-specific denominators,
   including combinations absent from the file. Compare with the planned
   design without assuming the planned counts are observed.
2. Define eligible RTs as positive finite times on correct trials. Report
   missing, non-finite, and non-positive RTs separately. Keep other extreme
   values; no protocol-based cutoff is known. Calculate accuracy from all
   trials with observed correctness, independently of RT eligibility.
3. Report pooled correct-trial RT mean, median, SD, and IQR by condition,
   with contributing counts. Also report equally weighted participant
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   accuracies and pooled trial accuracy, clearly distinguishing them.
4. Plot eligible trial RT distributions using faceted histograms with a
   shared 50-ms bin width and common axes. Show density on the y-axis.
   Note that the bin width is a display choice that can be varied.
5. Calculate each participant's median eligible RT and accuracy. Join by
   Subject to retain pairing. Make separate connected-point plots for RT
   and accuracy, using only complete pairs for the relevant outcome.
   Report complete and excluded pair counts. Use readable units, condition
   labels, and colour plus shape; do not add inferential error bars.
6. Summarise B-minus-A differences: milliseconds for participant RT medians
   and percentage points for accuracy. Describe the quantity averaged.
7. Compare the correct-only RT result with an all-response RT description
   using the same paired participants and the same valid-time rule. Explain
   that this changes the response subset, not merely the plotting method.
8. Verify one participant-condition RT median and accuracy directly from
   the imported data, using a separate base-R calculation. Handle missing
   outcomes, absent conditions, and no complete pairs explicitly.

Explain each output and its limitations. Keep calculated numerical
statements tied to R objects. Avoid causal claims, formal tests, and
unsupported declarations of a speed-accuracy trade-off. End with concise
guidance for interpreting the results and a transparent AI-assistance note.
```

The prompt is specific about the analysis while remaining honest about what has not been verified. It also requests checks that can reveal problems in the response. A focused follow-up prompt, if a check fails, would be:

> The participant–condition audit shows [the discrepancy]. Your calculation uses [the current denominator or grouping], whereas our plan specifies [the intended definition]. Explain the difference, correct the relevant code, and identify which figures, tables, and statements must be updated.

Example AI response — Complete Rmd source

See files:

- 04_student_activity2_reaction_time_REPORT.html
- 04_student_activity2_reaction_time_REPORT.Rmd

Review the worked response

Before adopting this example, check that its coding assumptions match your file and that its analytical choices match your question. In particular, inspect the accuracy denominators, the participant sets retained for each paired comparison, and the decision to condition the primary RT outcome on correctness. A successful run is necessary, but it is not sufficient validation of the report.

If your group's approach differs, explain what it estimates and compare results using a shared definition before deciding which approach is more useful. The value of the exercise lies in connecting statistical reasoning, visualization, reproducible computation, and critical evaluation of AI assistance.

Note on AI assistance

This feedback document and its worked Rmd example were prepared with assistance from ChatGPT (OpenAI). They provide general methodological commentary, not feedback derived from individual student submissions.